

Induction-Recursion for Self-Modifying Genetic Programs

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Abstract

We apply the theory of induction-recursion to the problem of genetic programs that modify their own syntax during evolution. Our framework guarantees termination of all evolved programs while still permitting a rich space of self-modifications. We prove that the class of expressible programs is equivalent to functions definable in Martin-Lof type theory with one universe, providing a precise characterization of the computational power of self-modifying GP and ruling out certain classes of undecidable behaviour.